Department of Computer Science and Engineering
Academic Year 2021 – 2022(~~Odd /~~ Even Semester)

Degree, Semester & Branch: B.E, VI & CSE
Course Code & Title: CS8603 & Distributed Systems
Name of the Faculty member (s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic: Unit III /** Deadlock detection in distributed systems: Introduction, System model , Preliminaries
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen: Role Play**
- **Justification:**

  The discovery of deadlocks in distributed systems is a difficult task. Because no site has a complete picture of the system's present status, and every inter-site connection has a finite and unpredictable delay. Because there is no global state maintenance in distributed systems, deadlock prevention (i.e., processes acquire essential resources before they begin execution) and avoidance (i.e., resources are granted to a process if the resulting global system state is secure) are impracticable. This active learning strategy is chosen to help students comprehend basic concepts like as deadlock avoidance, which they acquired in OS, and it will teach them about the impracticality of deadlock prevention implementation in distributed systems.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

  The course instructor remembered the notion of deadlock avoidance in operating systems. The loan was provided by Miss.Mutharuna, III CSE. Her bank account had the amount of Rs. 300. She delivered Rs. 100 to another student, Ms. Vaishnavi Rajam, and wrote down the facts on the white board about the sum she had and the bank's safety status. Ms. Jayarani, another loan borrower, received Rs.150 from the loan provider. In the white board, the loan provider noted the status of money and the safer state. Another borrower, Ms.V.S.Vaishanvi, has requested Rs.100 from the lender. However, the loan source may not be able to provide enough money. She marked her safer state as Unsafe state in the white board. Now, the course instructor explained the concept of deadlock avoidance with the safe and unsafe condition with the global state marked in the whiteboard by loan provider. The following figures shows the illustration of the activity.

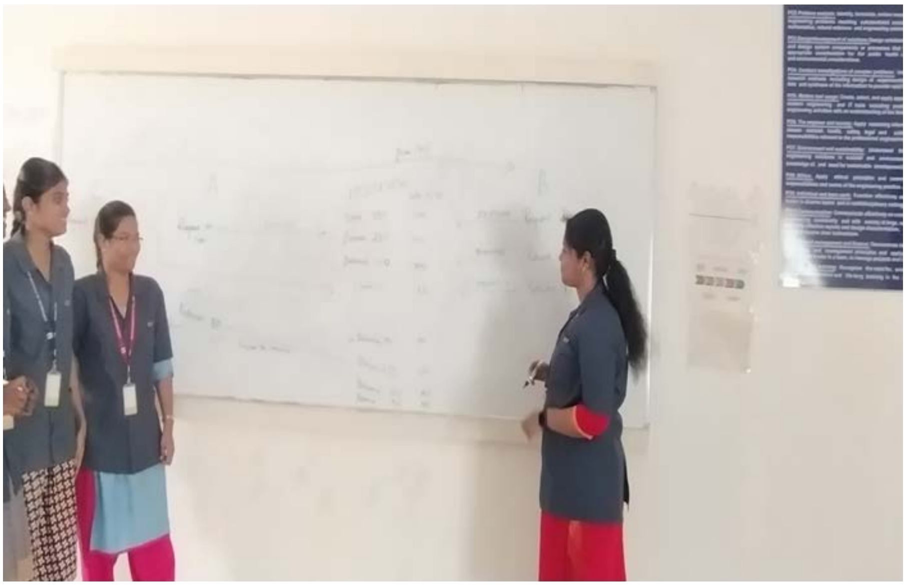

Fig.1. Ms. Vaishnavirajam requested loan amount to Loan Provider (Ms.Muthuaruna) and updated in global state of bank in white board.

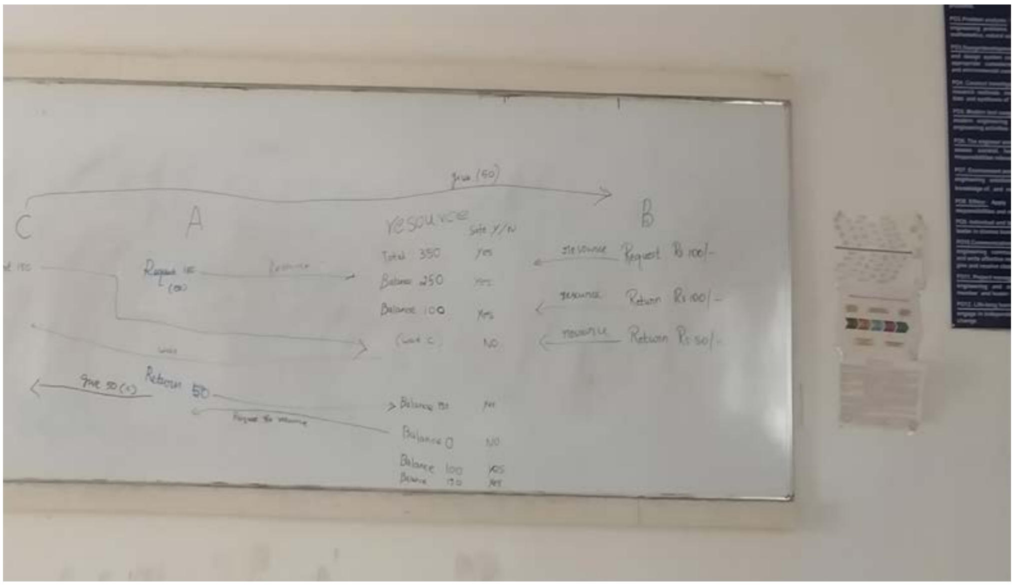

Fig.2. Global State of the Transaction details in Whiteboard

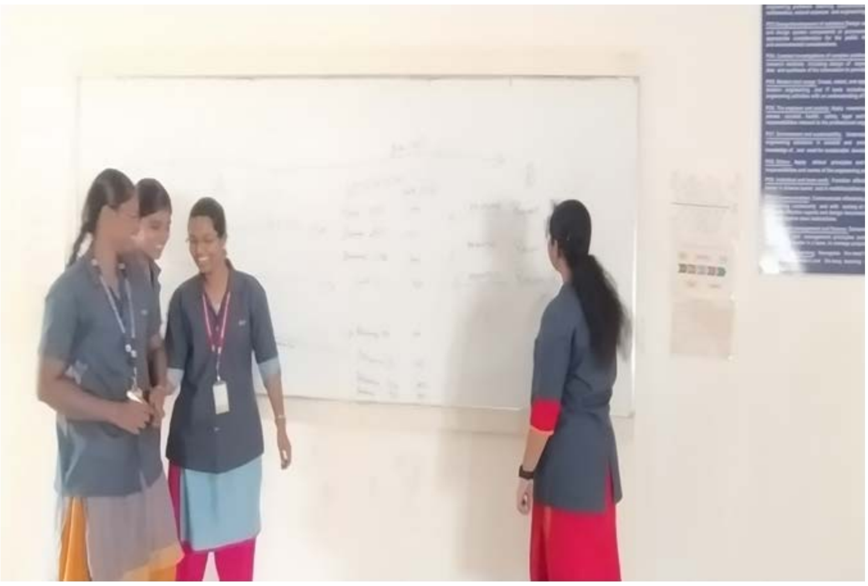

Fig.3. Loan provider (Miss. Muthuaruna) give money to Loan Requester (Miss. Jeyarani)

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 2	3	2	1	1	-	-	-	2	-	2	-	2	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 8**
	3	2	1	**1**	**2**
Justification for correlation	Students could be able to apply basic engineering theory and principles of Snapshot algorithm	Students could be identify their own example to explain the algorithm	Students could be able to recollect the basic principles of Snapshot algorithm	Students could able to interpret and explain the snapshot algorithm	Students interacted and shared their views about the concept with their peer group by ethically

Innovative practice	**PO 10**	**PO 12**	**PSO1**	**PSO 2**	**PSO 3**
	2	**2**	**3**	**3**	**2**
Justification for correlation	Students could be able to	Students could be able to learn the basic	Students could be able to use the snapshot	Students could be able to apply Global state in AI	Students could be able to use global state concept in

	communicate with their peer members and also they shared their views about the concept in front of the class to avoid their stage fear.	engineering concept to understand the working of snapshot algorithm	algorithm for implementing global state in the mobile computing technologies		IOT networks.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students felt that this activity is very useful and got good idea about the snapshot algorithm
  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - Most of the pairs of students explained the step by step procedure of snapshot algorithm in correct way of their own example.
    - Students interaction with their peer members and the class during the activity was good and effective.
  - ***Challenges faced in implementation:***
    - Due to time limitation, only few pairs of students were made to do the presentation in front of the class by random call.

References:

- https://www.kent.edu/ctl/think-pair-share
- Johnson, D., Johnson, R. (1999). Making Cooperative Learning Work: Theory into Practice, 38 (2), 67-73. Fitzgerald, D. (2013). Employing think-pair-share in associate degree nursing curriculum. Teaching & Learning in Nursing. 8, 3, p 88-90.
- Goodwin, M. (1999). Cooperative learning and social skills: What skills to teach and how to teach them. Intervention in School & Clinic, 35 (1), 29.
- Lyman, F. (1981). The responsive classroom discussions: the inclusion of all students. A. Anderson (Ed.), Mainstreaming Digest, College Park: University of Maryland Press, pp. 109-113.
- Mr.R.Venkatesh, "Introduction to Yoga and Meditation for professional excellence and stress management " in course GE 6075 Professional Ethics in Engineering, https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf

Signature of Faculty Member **HOD**